PT MERDEKA TSINGSHAN INDONESIA

STANDARD OPERATING PROCEDURE

LEAD AND ZINC RECOVERY POST

铅锌回收岗位

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1. Purpose 目的

This procedure covers the operator's responsibility for visual inspection, start up and shutdown of the lead and zinc recovery post under complete closure.

此程序涵盖操作员对完全关闭下的铅锌回收岗位进行目视操作前检查、启动运行和停车的责任。

2. Functional Description 功能说明

It is the responsibility of the operator to ensure that the lead and zinc recovery post is visually inspected before and after each shift to determine whether it can start and operate safely, or whether other activities (such as maintenance and repair or other operator tasks) must be performed before starting, to determine the sequential operation method of the lead and zinc recovery post and smooth operation, Stop order of normal stops and emergency stops.

操作员有责任确保在从停机启动之前和每个班次开始时对铅锌回收岗位进行目视检查，确定是否可以启动并安全运行，或者是否必须在开始启动之前执行其他活动(例如维护修理或其他操作员任务)，确定铅锌回收岗位的按照顺序启动的操作方法及平稳运行，正常停车和紧急停车的停止顺序。

3. References 参考

- MTI-CP-SOP-Post Technical Operation Procedures. MTI-CP-SOP岗位操作规程
- Solution Tank After Iron Removal(4402-TNK-013). 除铁后液槽(4402-TNK-013)
- Solution Pump After Iron Removal(4402-PSO-009/010). 除铁后液泵(4402-PSO-009/010)
- Zinc Removal Tank(4402-TNK-014/015). 沉锌槽(4402-TNK-014/015)
- Zinc Removal Pump(4402-PLS-009/010). 沉锌泵(4402-PLS-009/010)
- Zinc Filter Press(4402-FIL-005/006). 沉锌压滤机(4402-FIL-005/006)
- Washing Water Tank(4402-TNK-020). 洗涤水槽(4402-TNK-020)
- Washing Water Pump(4402-PSO-018/019). 洗涤水泵(4402-PSO-018/019)
- Belt Conveyor(4402-BCV-006/007). 皮带机(4402-BCV-006/007)
- Screwed Conveyor(4402-CNV-005/006). 螺旋输送机(4402-CNV-005/006)
- Zinc Precipitation Ton Bag Packer(4402-BPR-005/006). 沉锌吨袋包装机(4402-BPR-005/006)

4. PPE 个人保护装备



耳塞　安全帽　眼镜　劳保鞋　手套　工作服

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半面罩　　3M口罩

5. Hazards & Control 危害与控制

Description of Hazard / Environmental Issue 危害	Control Strategy 控制
1. Mechanical injury; 2. Liquid corrosion; 3. Electric shock; 4. Slippery conditions and a trip to the danger; 5. Asphyxia; 6. Vehicle collision crush; 7. Object falling from high altitudes; 8. Liquid sputtering; 9. Noise. 1. 机械伤害； 2. 液体腐蚀； 3. 触电； 4. 湿滑的条件，绊倒危险； 5. 窒息； 6. 车辆碰撞碾压； 7. 高空坠物； 8. 液体喷溅； 9. 噪音。	1. Install a protective cover on the rotating part of the equipment; 2. Wear appropriate labor protection supplies, once the skin contact with hydrochloric acid solution, immediately rinse with a lot of water; 3. Do not contact the electrical switch with wet hands during operation,Regular inspection of the equipment leakage protection device is complete, effective; 4. Please hold the handrail up and down the stairs; 5. Clearclearance or maintenance operations; 6. Walk along the prescribed route, and stay away from the lead zinc slag transport vehicles; 7. Wear labor protection goods and check the safety of stacking items on the operation platform; 8. Wear protective masks and gloves, as well as other protective products as required; 9. Wear the earplugs to protect your hearing. 1. 设备的转动部位安装防护罩； 2. 穿戴适宜的劳保用品，一旦皮肤接触到盐酸溶液，立即用大量清水冲洗； 3. 操作时不要湿手接触电器开关，定期检查设备漏电保护装置齐全、有效； 4. 上下楼梯请扶好扶手； 5. 清仓或维修作业执行审批程序； 6. 按照规定的路线行走，远离铅锌渣运输车辆； 7. 穿戴好劳保护品，检查各平台的物品堆放

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Description of Hazard / Environmental Issue 危害	Control Strategy 控制
	安全性； 8. 按要求戴好防护面具和防护手套，以及其它劳保护品； 9. 佩戴好耳塞，保护听力。

6. Equipment 工具/ 设备

4402-AGT-009/010	Zinc Removal Tank Agitator沉锌槽搅拌器
	Clean the site tools清扫现场工具

7. Procedure 程序

№	Stage	Key Points
1	General Check 普查	1. Confirm the pre-job technical training; 2. Wear suitable labor protection products, check and confirm that there is no unsafe state or safety risks on the site; 3. Check and clean the equipment and the site sanitation; 4. Contact with the process before and after, and give clear instructions. 1. 确认经过岗前技术培训； 2. 穿戴好适合的劳保护品，检查确认现场没有存在不安全状态或安全隐患； 3. 检查并打扫设备、现场卫生； 4. 与前后工序取得联系，并发出明确指令。
2	Inspect Solution Tank After Iron Removal (4402-TNK-013), Zinc Removal Tank (4402-TNK-014/015), Washing Water Tank (4402-TNK-020) 检查除铁后液槽(4402-TNK-013)，沉锌槽(4402-TNK-014/015)，洗涤水槽(4402-TNK-020)	1. Check whether the inlet and outlet pipelines and auxiliary pipelines, whether the installation of flange and valve meet the requirements, and whether the slurry storage tank leaks liquid; 2. Check the level control assembly is normal. 1. 检查出入口管线及附属管线，法兰、阀门安装是否符合要求，储浆槽是否漏液； 2. 检查液位控制组件是否正常。
3	Inspect Solution Pump After Iron Removal (4402-PSO-009/010), Zinc Removal Pump (4402-PLS-009/010),	1. Check whether the pump inlet pipeline and auxiliary pipeline, flange and valve installation meet the requirements, whether the anchor bolts and ground wire are good, and whether the coupling is installed; 2. Turn the pump shaft 3 times, should be flexible without

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	Washing Water Pump (4402-PSO-018/019) 检查除铁后液泵 (4402-PSO-009/010)，沉锌泵 (4402-PLS-009/010)，洗涤水泵(4402-PSO-018/019)	block; 3. Check whether the lubricating oil level is normal. The oil level should be between 1/2 and 2/3 of the oil lens; 4. Check the sealing condition of the pump is normal; 5. Point to move to confirm the correct steering; 6. Check whether the voltage meter, signal lamp and other meters of the pump control cabinet are normal; 7. Check whether the pump unit has air, if so, should be excluded; 8. Check that the pump head is under the liquid levels of the various tanks. 1. 检查泵出入口管线及附属管线，法兰、阀门安装是否符合要求，地脚螺栓及地线是否良好，联轴器及防护罩是否装好; 2. 手转动泵轴3圈，应灵活无阻滞; 3. 检查润滑油油位是否正常，油位应在油镜的1/2 ~ 2/3 之间; 4. 检查泵的密封状况是否正常; 5. 点动确认转向正确; 6. 检查泵控制柜电压表、信号灯等仪表指示是否正常; 7. 检查泵机组是否有空气，如有应予以排除; 8. 检查泵头是否处于各种槽的液位下。
4	Inspect Zinc Filter Press (4402-FIL-005/006) 检查沉锌压滤机 (4402-FIL-005/006)	1. Check whether the number of filter plates is sufficient, whether there is damage, whether the filter plate is clean, and whether the placement meets the requirements; 2. Check whether the filter cloth is folded, whether it is damaged, and whether the filter performance is good; 3. Check whether the pressure gauge of the mine pipeline is normal; 4. Check whether the connection is close and whether there is leakage; 5. Check whether the oil of the filter press is sufficient and whether the oil level meets the requirements (1 / 2~1 / 3); 6. Check whether the hydraulic oil top device is normal; 7. Check whether the emergency stop device is normal; 8. Check whether the pressure gauge of the hydraulic oil is normal and whether the pointer is in the specified interval; 9. Check whether the import and export of the filter press are smooth;

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№	Stage	Key Points
		10. Automatic plate drawing device; 11. Whether the water connection plate is flexible and easy to use. 1. 检查滤板数量是否足够，有无破损，滤板是否清洁，安放是否符合要求； 2. 检查滤布是否折叠，有无破损，过滤性能是否良好； 3. 检查给矿管路压力表是否正常； 4. 检查各处联接是否紧密，有无泄露； 5. 检查压滤机油是否足够，油位是否符合要求(1/2 ~ 1/3)； 6. 检查液压油顶装置是否正常； 7. 检查急停装置是否正常； 8. 检查液压油的压力表是否正常，指针是否在规定的区间； 9. 检查压滤机各进出口是否畅通； 10. 检查自动拉板装置良好； 11. 接水盘是否灵活好用。
5	Inspect Belt Conveyor(4402-BCV-006/007) 检查皮带机(4402-BCV-006/007)	1. Each bearing is well lubrication and closed, the reducer oil level is appropriate, the upper and lower roller lubrication is good, the operation is flexible, whether the wear of the first and tail wheels and the tape are filled with materials, and whether the cleaning device of the first and tail wheels is in good condition; 2. Comprehensively check whether all parts of the belt conveyors are intact, whether the tension of the tape is appropriate, and whether the corresponding tape tightening device and dust-proof facilities are in good condition; 3. Whether the electrical facilities and control system are intact, whether the protective device is firm, and whether the funnel baffle and liner are damaged; 4. Check whether the cushion rollers of the impact load parts are intact; 5. Check the installation of belt sweeper; 6. Whether the installed electrical facilities and signal devices are intact and responsive; 7. Place the field control box in the manual position, press the start button, slowly rotate the potentiometer, and watch whether the running direction of the tape is correct, and whether the belt can run away; 8. After checking and contacting the post, press the start

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№	Stage	Key Points
		button; 9. During the operation process, check whether the tape machine emergency stop device, belt deviation correction and speed measuring sensor are in good condition. 1. 各部轴承润滑与封闭良好，减速机油位适当，上下托滚润滑良好，运转灵活，首、尾轮磨损情况与胶带间是否夹有物料，首、尾轮的清扫装置是否完好； 2. 全面检查皮带机各部分是否完好无损，胶带的张紧度是否合适，相应的胶带拉紧装置及防尘设施是否完好； 3. 电器设施和控制系统是否完好，防护装置是否牢固，漏斗挡板和衬板是否损坏； 4. 检查承受冲击载荷部位的缓冲托辊是否完好； 5. 检查皮带机清扫器安装是否正常； 6. 装设的电器设施及信号装置是否完好和反应灵敏； 7. 将现场控制盒置于手动位置，按一下启动按钮，迟缓旋动电位器，观看胶带运转方向能否正确，皮带是否跑偏； 8. 检查无误并与前后岗位联系后，按下启动按钮； 9. 运转过程中，检查皮带机急停装置、皮带纠偏和测速传感器是否控制灵敏。
6	Inspect Screwed Conveyor (4402-CNV-005/006) 检查螺旋输送机(4402-CNV-005/006)	1. Check whether the lubricating grease in the bearing is appropriate; 2. Check whether the oil quality and quantity of the lubricating oil in the reducer are normal; 3. Check whether the bolts are fastened; 4. Check whether the cover plate of the screw conveyor must be tightly covered and whether the conveyor is clean; 5. Check whether the spiral blade is deformed ; 6. Determine whether the rotation direction of the motor meets the work requirements. 1. 检查轴承内润滑脂是否合适； 2. 检查减速机内的润滑油油质、油量是否正常； 3. 检查各螺栓是否紧固； 4. 检查螺旋输送机的盖板必须盖严，输送机内是否清理干净； 5. 检查螺旋叶片是否变形； 6. 判明电机旋转方向是否符合工作要求。

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№	Stage	Key Points
7	Inspect Zinc Precipitation Ton Bag Packer(4402-BPR-005/006) 检查沉锌吨袋包装机(4402-BPR-005/006)	a. Check whether the safety shield of each transmission part is intact and effective; b. Check whether the lubrication part has been grease; c. Check whether the indicator instrument is working normally, and the signal is sensitive and effective; d. Check whether the material scraping and fixing screw is loose; e. Check whether the pneumatic butterfly valve is normal and flexible; f. Check whether the lifting scale body platform is normal; g. Check whether the automatic uncoupling device is normal; h. Check whether the pneumatic clip is normal; i. Check the control system for normal operation. a. 检查各传动部位安全防护罩是否完好有效; b. 检查需要润滑部位是否已加油脂; c. 检查指示仪表工作是否正常，信号灵敏有效; d. 检查料刮固定螺丝是否松动; e. 检查气动蝶阀是否正常、灵活; f. 检查升降秤体平台是否正常; g. 检查自动脱钩装置是否正常; h. 检查气动夹袋器是否正常; i. 检查控制系统是否正常。
8	Check the other Equipment and Facilities 检查其它设备设施	1. Zinc Removal Tank Agitator(4402-AGT-009/010) a. Check whether the screws of the mixing paddle are fastened; b. Check whether the positive and reverse version of the agitator is correct; c. Check the surface of the mixer is smooth, without rough and pitting phenomenon; d. Check whether the mechanical seal of the mixer is good and whether the oil seal is normal; e. Electric mixer idling, check for abnormal sound, whether the up and down are concentric or shaking. 1. 沉锌槽搅拌器(4402-AGT-009/010) a. 检查搅拌桨的螺丝是否紧固; b. 检查搅拌器的正反转是否正确; c. 检查搅拌器的搅拌桨表面光滑，无粗糙和点蚀现象; d. 检查搅拌器的机械密封是否良好，各油封是否正常;

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№	Stage	Key Points
		e. 电动搅拌器进行空转，检查是否有异常声响，上下是否同心或摇晃。 2. Chute a. Check zinc filter press (4402-FIL-005/006) whether the feeding chute is in good condition; b. Check zinc filter press (4402-FIL-005/006) whether the blocking detection switch of the feeding chute is normal. 2. 溜槽 a. 检查沉锌压滤机(4402-FIL-005/006)下料溜槽是否完好； b. 检查沉锌压滤机(4402-FIL-005/006)下料溜槽的堵料检测开关是否正常。 3. Valve a. Check whether the butterfly valve (100BF) at the bottom of the removal tank (4402-TNK-013) is normal and flexible; b. Check whether the three butterfly valves (300BF, 300BF, 300BF) at the inlet (4402-PSO-009/010) and the two butterfly valves (250BF, 250BF) are normal and flexible; c. Check the two check valves (100CH, 100CH) from the lime milk conveying pump (4402-TNK-014/015) (4524-PSL-003/004), the two butterfly valves (125BF, 125BF) pumped by the zinc sink pump (4402-PLS-009/010), and the two butterfly valves (100BF, 100BF, 100 BF) are normal and flexible; d. Check whether two butterfly valves (200BF, 200BF) at the inlet of zinc sinking pump (4402-PLS-009/010) and two butterfly valves (125BF, 125BF) are normal and flexible; e. Check the zinc sink filter (4402-FIL-005/006) washing pump (4402-PSO-018/019) four butterfly valves (100BF, 100BF, 100BF, 100BF), Two butterfly valves (125BF, 125BF) pumped from the zinc sink pump (4402-PLS-009/010), Four ball valves with compressed air (4601-CAT-001/002/003/004) (50BA, 50BA, 100BA, 100BA), Two butterfly valves (100BF, 100BF), whether the two butterfly valves (100BF, 100BF) are normal and flexible after washing; f. Check the washing sink (4402-TNK-020) new water

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№	Stage	Key Points
		a gate valve (50GA), the bottom discharge of a butterfly valve (100BF) is normal and flexible; g. Check whether two butterfly valves (125BF) at the inlet of the washing pump (4402-PSO-018/019) and the two butterfly valves (100BF) are normal and flexible. 3. 阀门 a. 检查除铁后液槽(4402-TNK-013)底部排液一个蝶阀(100BF)是否正常、灵活; b. 检查除铁后液泵(4402-PSO-009/010)入口三个蝶阀(300BF,300BF,300BF)，出口两个蝶阀(250BF,250BF)是否正常、灵活; c. 检查沉锌槽(4402-TNK-014/015)的来自石灰乳输送泵(4524-PSL-003/004)的两个止回阀(100CH,100CH)，沉锌泵(4402-PLS-009/010) 泵过来的两个蝶阀(125BF,125BF)，底部排液两个蝶阀(100BF,100BF)是否正常、灵活; d. 检查沉锌泵(4402-PLS-009/010)入口三个蝶阀(200BF,200BF,200BF)，出口两个蝶阀(125BF,125BF)是否正常、灵活; e. 检查沉锌压滤机(4402-FIL-005/006)洗涤水泵(4402-PSO-018/019泵过来的洗涤水蝶阀四个蝶阀(100BF,100BF,100BF,100BF)，沉锌泵(4402-PLS-009/010)泵过来的两个蝶阀(125BF,125BF)，压缩空气(4601-CAT-001/002/003/004)的四个球阀(50BA,50BA,100BA,100BA)，滤液两个蝶阀(100BF,100BF)，洗涤后液两个蝶阀(100BF,100BF)是否正常、灵活; f. 检查洗涤水槽(4402-TNK-020)新水一个闸阀(50GA)，底部排液一个蝶阀(100BF)是否正常、灵活; g. 检查洗涤水泵(4402-PSO-018/019)入口两个蝶阀(125BF,125BF)，出口两个蝶阀(100BF,100BF)是否正常、灵活。

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№	Stage	Key Points
9	Inspect Materials Level 检查物料料位	1. Check whether the liquid level of the solution tank after iron removal (4402-TNK-013), zinc removal tank (4402-TNK-014/015), washing water tank (4402-TNK-020) meets the pressure requirements; 2. Check whether the material is stored in the zinc filter press (4402-FIL-005/006), whether the hydraulic oil is sufficient, and whether there is a filter cake in the feeding chute; 3. Check whether the material is stored on the tape machine(4402-BCV-006/007), and there is no adhesive deposit on the surface of the head and tail wheel and the belt joint; 4. Check whether the material is stored in the screw conveyor (4402-CNV-005/006). 1. 检查除铁后液槽(4402-TNK-013)、沉锌槽(4402-TNK-014/015)和洗涤水槽(4402-TNK-020)液位是否达到要求; 2. 检查沉锌压滤机(4402-FIL-005/006)内是否存料，液压油是否足够，下料溜槽是否存有滤饼; 3. 检查皮带机(4402-BCV-006/007)上是否存料，首尾轮表面与皮带结合部无黏附积矿存在; 4. 检查螺旋输送机(4402-CNV-005/006)内是否存料。
10	Start-Up Of The Lead And Zinc Recovery Post 铅锌回收岗位开机	1. No-load start should be guaranteed when starting; 2. The driving sequence is along the material direction, from the back to drive that is, the reverse driving, taking into account to reduce the idling of the equipment, each equipment has been delayed to open; At the same time, Confirm that the process behind the lead and zinc recovery post has been started and has the feeding conditions: Open the washing sink (4402-TNK-020) new water a gate valve (50GA), and close the new water ⟹ Open the PP valve (***) from the drain tank (4402-TNK-013) and fill the drain fluid, Start the zinc sink tank agitator (4402-AGT-009/010), Open the iron removal pump (4402-PSO-009/010)) outlet two butterfly valves (250BF, 250BF), Start the pump (4402-PSO-009/010), Open the iron removal pump (4402-PSO-009/010) inlet three butterfly valves (300BF, 300BF, 300BF), Pump the iron drain fluid into the zinc sink (4402-TNK-014/015) ⟹ Inform the lime milk preparation post to open the two check valves (100 CH, 100 CH) of the lime milk delivery pump

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№	Stage	Key Points
		(4524-PSL-003/004), Start the lime milk conveying pump (4524-PSL-003/004) and transport the lime milk to the zinc sink tank (4402-TNK-014/015)→ Start zinc sink filter press (4402-FIL-005/006)↳After the zinc sinking, press the "Start" button of sinking zinc filter (4402-FIL-005/006), The oil pump starts to work, and then press the "press" button, the piston pushes the pressing plate to press, when the pressure force reaches the high pressure, the hydraulic system will automatically stop and maintain the pressure, open the filter inlet butterfly valve (125BF,125BF) and the filter discharge liquid butterfly valve (100BF,100BF)→ Open two butterfly valves (125BF, 125BF) of zinc sink pump (4402-PLS-009/010), Starting the zinc sinking pump (4402-PSL-009/010), Open three butterfly valves (200BF, 200BF, 200BF) at the moving zinc pump (4402-PSL-009/010) → The slurry is put into each filter chamber through the feed hole on the thrust plate, and then filtered under the feed pressure not greater than 0.6MPa to gradually form the filter cake→ After the filter press filter cake is filled, Close the three butterfly valves of the pump pump (4402-PSL-009/010) (200BF,200BF, 200BF), Stop the zinc sinking pump (4402-PSL-009/010), Close two butterfly valves (125BF, 125BF) (4402-PSL-009/010) and the filter press inlet butterfly valve (125BF)→ Open the washing water pump (4402-PSO-018/019) outlet for two butterfly valves (100BF, 100BF), Start the washing water pump (4402-PSO-018/019), Open two butterfly valves (125BF, 125BF) of washing pump (4402-PSO-018/019), Wash the filter cake at a pressure not greater than 0.6MPa. After the operation, close outlet of the two butterfly valves (100BF, 100BF)→Four ball valves (50BA, 50BA, 100BA, 100BA) slowly opening compressed air (4601-CAT-001/002/003/004), Dry the filter cake at a pressure not greater than 0.6MPa, and close the compressed air ball valve (50BA, 50BA, 100BA, 100BA) after the operation→Press and hold the "release" button on the operation panel, and the pressing plate will automatically stop after it is sent back in place, and at the same time, the pull plate hook moves forward to unload the reciprocating pull plate↳→During the filter cake unloading process, Start spiral conveyor (4402-CNV-005/006)→ Start the lead and zinc ton bag packing machine (4402-BPR-005 / 006)→ After the packing machine, start the belt machine (4402-BCV-

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№	Stage	Key Points
		006 / 007), and the forklift fork takes the ton bag. 1. 启动时应保证空载启动； 2. 开车顺序为沿物料走向，从后往前开即逆向开车，兼顾减少设备的空转，各设备相继延时开启；同时，确认铅锌回收岗位后面的工序已经开车并具备给料条件： 打开洗涤水槽(4402-TNK-020)新水一个闸阀(50GA)，并注满新水后关闭⟶打开待除铁后液槽(4402-TNK-013)的PP过来的阀门(***)，充满除铁后液后，启动沉锌槽搅拌器(4402-AGT-009/010)，打开除铁后液泵(4402-PSO-009/010))出口两个蝶阀(250BF,250BF)，启动除铁后液泵(4402-PSO-009/010)，打开除铁后液泵(4402-PSO-009/010)入口三个蝶阀(300BF,300BF,300BF)，将除铁后液泵入沉锌槽(4402-TNK-014/015)⟶通知石灰乳制备岗位打开石灰乳输送泵(4524-PSL-003/004)的两个止回阀(100CH,100CH)，启动石灰乳输送泵(4524-PSL-003/004)，输送石灰乳至沉锌槽(4402-TNK-014/015)⟶启动沉锌压滤机(4402-FIL-005/006)沉锌结束后，按下沉锌压滤机(4402-FIL-005/006)“启动”按钮，油泵开始工作，然后再按一下“压紧”按钮，活塞推动压紧板压紧，当压紧力到达调定高点压力后，液压系统自动跳停并保持压力，打开压滤机入口蝶阀(125BF,125BF)和压滤机排液蝶阀(100BF,100BF)⟶打开沉锌泵(4402-PLS-009/010)出口两个蝶阀(125BF,125BF)，启动沉锌泵(4402-PSL-009/010)，打开沉锌泵(4402-PSL-009/010)入口三个蝶阀(200BF,200BF,200BF)⟶料浆通过止推板上的进料孔进去各个滤室，在进料压力不大于0.6MPa下加压过滤，逐渐形成滤饼⟶压滤机滤饼充满后，关闭沉锌泵(4402-PSL-009/010)入口三个蝶阀(200BF,200BF,200BF)，停止沉锌泵(4402-PSL-009/010)，关闭沉锌泵(4402-PSL-009/010)出口两个蝶阀(125BF,125BF)和压滤机入口蝶阀(125BF)⟶打开洗涤水泵(4402-PSO-018/019)出口两个蝶阀

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		(100BF,100BF)，启动洗涤水泵(4402-PSO-018/019)，打开洗涤水泵(4402-PSO-018/019)入口两个蝶阀(125BF,125BF)，在压力不大于0.6MPa下对滤饼进行洗涤，操作完成后，关闭洗涤水出口两个蝶阀(100BF,100BF)⟹缓慢打开压缩空气(4601-CAT-001/002/003/004)的四个球阀(50BA,50BA,100BA,100BA)，在压力不大于0.6MPa下对滤饼进行风干，操作完成后，关闭压缩空气球阀(50BA, 50BA, 100BA, 100BA)⟹按住操作面板上的“松开”按钮，压板退回到位后会自动停止，同时拉板钩前进进行往复拉板卸料⟹滤饼卸料过程中，启动螺旋输送机(4402-CNV-005/006)⟹启动铅锌吨袋包装机(4402-BPR-005/006)⟹吨袋包装机打包结束后启动皮带机(4402-BCV-006/007)，叉车叉走吨袋。
11	The Lead And Zinc Recovery Post Stops 铅锌回收岗位停机	1. Orderly shutdown a. The materials in the standby machine should be discharged before stopping; b. Stop in reverse order of driving order and material discharge order, Each equipment needs to stop: After notifying the lime milk preparation post to stop the lime milk delivery pump (4524-PSL-003/004), Close the two check valves (100CH, 100CH) of the lime milk delivery pump (4524-PSL-003/004)⟹After the removal tank (4402-TNK-013) is emptied, Close the three butterfly valves (300BF, 300BF,300BF) of the pump (4402-PSO-009/010), Stop the pump (4402-PSO-009/010), Close removal pump (4402-PSO-009/010) outlet two butterfly valves (250BF, 250BF), Stop the zinc sink tank agitator (4402-AGT-009/010)⟹After draining of the zinc drain tank (4402-TNK-014 / 015), Close the zinc sink pump (4402-PSL-009/010) inlet three butterfly valves (200BF, 200BF, 200BF), Stop the zinc sinking pump (4402-PSL-009/010), Close two butterfly valves (125BF, 125BF) of zinc pump (4402-PSL-009/010)⟹Close the filter press inlet butterfly valve (125BF), Press the "Stop" button of the sinking zinc filter press (4402-FIL-005/006) to turn off the power supply; Close the four ball valves (50BA, 50BA, 100BA, 100BA) of compressed air (4601-CAT-001/002/003/004)⟹ Stop spiral Conveyor (4402-CNV-005 / 006)⟹ Stop

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№	Stage	Key Points
		machine (4402-BPR-005 / 006)⟹ Stop belt conveyor (4402-BCV-006 / 007)⟹Clean the site for hygiene. 1. 正常停车 a. 停车时应待机内物料排净后再停车； b. 按照开车顺序逆序和物料排净顺序停车，各设备都需要延时停止： 通知石灰乳制备岗位停止石灰乳输送泵(4524-PSL-003/004)后，关闭石灰乳输送泵(4524-PSL-003/004)的两个止回阀(100CH,100CH)⟹**待除铁后液槽(4402-TNK-013)排空后**，关闭除铁后液泵(4402-PSO-009/010)入口三个蝶阀(300BF,300BF,300BF)，停止除铁后液泵(4402-PSO-009/010)，关闭除铁后液泵(4402-PSO-009/010)出口两个蝶阀(250BF,250BF)，停止沉锌槽搅拌器(4402-AGT-009/010)⟹**待沉锌槽(4402-TNK-014/015)排空后**，关闭沉锌泵(4402-PSL-009/010)入口三个蝶阀(200BF,200BF,200BF)，停止沉锌泵(4402-PSL-009/010)，关闭沉锌泵(4402-PSL-009/010)出口两个蝶阀(125BF,125BF)⟹关闭沉锌压滤机(4402-FIL-005/006)入口蝶阀(125BF)，按下压滤机“停止”按钮，关闭电源；关闭压缩空气(4601-CAT-001/002/003/004)的四个球阀(50BA,50BA,100BA,100BA)⟹停止螺旋输送机(4402-CNV-005/006)⟹停止沉锌吨袋包装机(4402-BPR-005/006)⟹停止皮带机(4402-BCV-006/007)⟹打扫现场卫生。 2. Emergency Stop a. When an emergency occurs, Close solution pump after iron removal (4402-PSO-009/010), zinc removal pump (4402-PLS-009/010), zinc filter press (4402-FIL-005/006), washing water pump (4402-PSO-018/019), belt conveyor (4402-BCV-006/007), screwed conveyor (4402-CNV-005/006), zinc precipitation ton bag packer (4402-BPR-005/006), zinc removal tank agitator(4402-AGT-009/010), After all stop the operation, Check whether the start / stop button of each device is in the stop state, Close zinc removal pump (4402-PLS-009/010) of three butterfly valves at the inlet (200BF, 200BF,

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№	Stage	Key Points
		200BF) and two butterfly valves at the outlet (125BF, 125BF), zinc filter press (4402-FIL-005/006) of inlet butterfly valve (125BF), compressed air (4601-CAT-001/002/003/004) of the four ball valves (50BA,50BA,100BA,100BA), washing water pump (4402-PSO-018/019) of two butterfly valves at the inlet (125BF, 125BF) and two butterfly valves (100BF, 100BF), lime pump (4524-PSL-003/004) of two check valves (100CH,100CH), solution pump after iron removal (4402-PSO-009/010) of three butterfly valves at the inlet (300BF, 300BF, 300BF) and two butterfly valves at the outlet (250BF, 250BF),, at the same time, the upper level of equipment all need to stop; b. After the accident, the equipment is restored to normal. If production still needs to be conducted, the operation shall be restarted in the driving sequence; c. After the accident treatment, the equipment returns to normal. If the production is not necessary, operate according to the normal parking order. 2. 紧急停车 a. 当出现紧急事故，关闭除铁后液泵(4402-PSO-009/010)，沉锌泵(4402-PLS-009/010)，沉锌压滤机(4402-FIL-005/006)，洗涤水泵(4402-PSO-018/019)，皮带机(4402-BCV-006/007)，螺旋输送机(4402-CNV-005/006)和沉锌吨袋包装机(4402-BPR-005/006)，沉锌槽搅拌器(4402-AGT-009/010)，全部停止运行后，检查各设备的启动/停止按钮是否处于停止状态，关闭沉锌泵(4402-PSL-009/010)入口三个蝶阀(200BF,200BF,200BF)和出口两个蝶阀(125BF,125BF)，压滤机入口蝶阀(125BF)，压缩空气(4601-CAT-001/002/003/004)的四个球阀(50BA,50BA,100BA,100BA)，洗涤水泵(4402-PSO-018/019)入口两个蝶阀(125BF,125BF)和出口两个蝶阀(100BF,100BF)，石灰乳输送泵(4524-PSL-003/004)的两个止回阀(100CH,100CH)，除铁后液泵(4402-PSO-009/010)入口三个蝶阀(300BF,300BF,300BF)和出口两个蝶阀(250BF,250BF)，同时，上一级设备均需停车；

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№	Stage	Key Points
		b. 事故处理后，设备恢复正常，如仍需要进行生产，按照开车顺序重新启动操作; c. 事故处理后，设备恢复正常，如不需要进行生产，按照正常停车顺序进行操作。
12	Permits &Isolation Sheets 许可证和隔离表	1. Only authorized isolators can sign off and cancel permits and registration sheets. 1. 只有经过授权的隔离者才能签署和取消许可证和登记表。

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8. Lead And Zinc Recovery Post 铅锌回收岗位 Questionnaire

Declaration of Understanding

I (_____Learner_____) have completed the theory assessment and had all my valid, reliable, and consistent with my level of knowledge in the assessment task.

Signature
Date:

I (_____Assessor_____) have witnessed the above learner complete the theory assessment and believe the assessment has provided a valid, reliable, and consistent result.

Signature
Date:

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9. Appendix

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Standard Operating Procedure
标准操作程序

Lead and zinc recovery posts in the chloride plant
CP 铅锌回收岗位

SOP Details 程序明细

Processing Department 氯化厂	Procedure Level 程序级别: **3**	Revision Number 修订编号: 0
Date Approved 审批日期:	Date Issued 颁布日期:	Next Review Date 下次审核日期:
Details of Revision 修订明细:		

Revision Details 修订明细

Rev. No 版本号	Date 日期	Section 章节	Page 页码	Revision Description 修订说明	Reviewed By 审核人
1					
2					
3					

Approvals 批准

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